

Double star

X17 (2017) — English translation

A binary star is a system of two gravitationally bound stars revolving around their center of mass in circular orbits. Cosmonaut-researcher Gluck wants to find such orbits that, when circling, his spacecraft would be on a straight line connecting these stars, and so that fuel consumption would be minimal. Use the following notations: L – distance between stars, $L_x = xL$ – distance from the spacecraft to the star M_1 , star mass ratio $k = M_1/M_2$.

A1	Express the parameter $k = k(x)$.	4.50
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A2	Estimate numerically at what distances from the Moon Gluck's ship must be in order for it to move in the Earth-Moon system as described in the problem statement. Consider the ratio of the masses of the Moon and the Earth to be $k = 1/81$, and the distance from the Earth to the Moon to be $L = 3.8 \cdot 10^5$ km.	3.00
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A3	Schematically depict the possible positions of the satellite in the Earth-Moon system.	0.50
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Source: <https://pho.rs/p/302>